

IB MATH AA SL

Lesson 40

Kinematics with Calculus

Calculus · 60-minute lesson

Syllabus SL 5.9 · Property of Lynda Badmus Education

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Do Now — recall

ON YOUR WHITEBOARD

1. Differentiate $s = 5t^2$.
2. What is velocity, in words?
3. What does acceleration measure?

Answers (click to reveal)

1. $10t$
2. rate of change of displacement
3. rate of change of velocity

What we're learning today

LEARNING OBJECTIVES

- 1 Relate**
displacement, velocity and acceleration.
- 2 Differentiate**
s to get v, and v to get a.
- 3 Find**
velocity and acceleration at a time t.
- 4 Interpret**
when an object is at rest ($v = 0$).

SUCCESS CRITERIA — I CAN...

- ✓
use $v = ds/dt$ and $a = dv/dt$
- ✓
find $v(t)$ and $a(t)$ by differentiating
- ✓
evaluate v and a at a given time
- ✓
find when the object is momentarily at rest

The chain of motion

DIFFERENTIATE DOWN

$$v = \frac{ds}{dt}, \quad a = \frac{dv}{dt}$$

s → v

differentiate displacement to get velocity

v → a

differentiate velocity to get acceleration

At rest

velocity $v = 0$

DIRECTIONS

Differentiate

$s \rightarrow v \rightarrow a$ (down)

Integrate

$a \rightarrow v \rightarrow s$ (up, next unit)

v=0

momentarily at rest / turning.

Velocity & acceleration

QUESTION

A particle has displacement $s = t^3 - 6t^2$ (metres).

Find v and a .

SOLUTION

$v = ds/dt$:

$$v = 3t^2 - 12t$$

$a = dv/dt$:

$$a = 6t - 12$$

Your turn — motion

ANSWER IN YOUR BOOK

1. $s = 4t^2$. Find v .
2. $v = 6t - 2$. Find a .
3. $s = t^3$. Find v and a .

Answers (click to reveal)

1. $v = 8t$
2. $a = 6$
3. $v = 3t^2$, $a = 6t$

At rest & evaluating

$$v = 0$$

$$v(t) = 0 \Rightarrow \textit{at rest}$$

At rest

solve $v(t) = 0$ for t

At a time

substitute t into $v(t)$ or $a(t)$

UNITS

s

metres (m)

v

m s^{-1}

a

m s^{-2}

When is it at rest?

QUESTION

With $v = 3t^2 - 12t$,
find when the particle is at rest.

SOLUTION

Set $v = 0$ and factor:

$$3t^2 - 12t = 3t(t - 4) = 0$$

So $t = 0$ or $t = 4$ seconds.

Exam-style question

A particle moves with $s = 2t^3 - 9t^2 + 12t$ (metres, $t \geq 0$).

(a) Find v . [2]

(b) Find the times when the particle is at rest. [3]

MARK SCHEME — reveal after students attempt (tutor only)

(a) $v = 6t^2 - 18t + 12$ (A1)(A1)

(b) $6t^2 - 18t + 12 = 0 \rightarrow t^2 - 3t + 2 = 0$ (M1) $\rightarrow (t-1)(t-2)=0$ (A1) $\rightarrow t = 1, 2$ s (A1)

Common mistakes



Direction

$s \rightarrow v \rightarrow a$ by differentiating; don't integrate by mistake.



At rest

'at rest' means $v = 0$, not $a = 0$.



Evaluate

Substitute t into the correct function (v or a).



Units

m , $m\ s^{-1}$, $m\ s^{-2}$ respectively.

Mixed practice

QUESTIONS

1. $s = 3t^2$. Find v .
2. $v = 4t - 8$. When at rest?
3. $s = t^3 - 3t$. Find v .
4. $v = 2t$. Find a .

Answers (click to reveal)

1. $v = 6t$
2. $t = 2$
3. $v = 3t^2 - 3$
4. $a = 2$

Plenary & exit ticket

KEY FORMULAS

$$v = \frac{ds}{dt}$$

$$a = \frac{dv}{dt}$$

EXIT TICKET — before you leave

1. $s = 5t^2$. Find v .
2. $v = 3t^2 - 3$. When at rest?
3. a from $v = 10t$?

Answers (click to reveal)

1. $v = 10t$
2. $t = \pm 1 \rightarrow t = 1$ ($t \geq 0$)
3. $a = 10$

Self-assess:
• need more help
• nearly there
• confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Kinematics with calculus.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 41

Integration & antiderivatives — reversing differentiation.

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$